Multimodal Emotional Recognition System

Paras Khanal, Vincent Sanchez, Shokhina Jalilova

New Mexico State University

Department of Computer Science

New Mexico State University

Introduction

- human communication increasingly occurs through online platforms such as social media, messaging applications, discussion forums, video conferencing systems, and digital review platforms.
- These interactions often contain rich emotional information expressed through written language, facial expressions, and other behavioral cues.
- At the same time, Artificial Intelligence (AI) has become deeply integrated into daily life

Applications

- Human-computer interaction
- Online education
- Customer satisfaction
- Healthcare
  - Mental health improvement
  - Behavior health
  - Telemedicine
  - Virtual care

Motivation

- Although modern AI systems can communicate with humans, they still struggle to truly understand human emotions.
- In facial image, lighting, face angles, similar facial expressions, cultural variation make it challenging
- Emotions are subtle and context-dependent
- Single-modality systems (text-only or image-only) often miss important emotional cues.
- These limitations reduce the ability of AI systems to recognize subtle emotional indicators associated with increasing mental health issues.

Motivation

- AI systems can communicate with humans, but they still struggle to understand human emotions.
- Emotional intelligence is essential for creating more natural, empathetic, and human-centered AI systems.
- Multimodal emotion recognition can improve applications in various fields, such as Mental health support, online education, human-computer interaction
  - Healthcare
  - Education
  - Customer service
  - Human-computer interaction
- This work aims to advance affective computing for real-world emotionally aware AI applications.

Related work

- Dr. Rosalind Picard pioneered through the field **Affective (Emotional) Computing** - published book in 1977

- She states the emotional intelligence as a **core component for intelligent machines.**

- Her applications include:
  - Autism and Affectiva: manage seizure risk and facilitate better understanding of emotions.
  - Precision Medicine for depression

- Her mission is to mitigate epidemic of depression and suicide – current rate: taking more lives and costing more long-term disability than cancer.

Rosalind W. Picard, Sc.D., FIEEE
Director of Affective Computing Research
MIT Media Lab, E14-348A
75 Amherst Street
Cambridge, MA 02139; USA

Problem Statement

We define emotion recognition as a supervised multi-class classification problem

- Input: Paired text and visual sample
  - x_t = text
  - x_v= facial image/frame
- Output: an emotion label
  - $\hat{y} \in Y$
- Model idea:
  - $f(x_t, x_v) \rightarrow \hat{y}$

- Text and image are encoded separately:
  - $h_t = f_t(x_t), h_v = f_v(x_v)$
- Features are then combined:
  - $h = concat(h_t, h_v)$
- Classify emotion using h:
  - $\hat{y} = classifier(h)$

Key idea: Combine language and facial cues to predict emotion more accurately than using one modality alone

Datasets

Twitter Emotions:

416,809 tweets, 6 classes (sadness, joy, love, anger, fear, surprise). Moderate imbalance: joy+sadness ~60% of samples. Used 80/20 stratified split.

OAHEGA (Facial):

~15,500 RGB images, 6 classes. Significant imbalance: Happy/Neutral/Sad ~2,700 each vs. Angry/Surprise/Ahegao ~900. Used 70/15/15 split.

MELD (Multimodal):

13,708 utterances from Friends TV, 7 classes, synchronized text + video. Severe imbalance: neutral 46.8%, disgust+fear only 3.5%. Pre-split train/dev/test.

Key challenge:

MELD neutral:disgust ratio up to 25:1 — accuracy alone is misleading. Cohen's Kappa required for honest evaluation.

Challenges

Class imbalance was a persistent challenge across all three datasets.

Initial approach:

- Raw inverse frequency weights $w_k = N/(K \times n_k)$ combined with WeightedRandomSampler. Result: catastrophic over-correction on OAHEGA — accuracy dropped from 55% to 20%, model predicted only minority classes.

Fix:

- Remove WeightedRandomSampler entirely. Replace with square-root inverse frequency: $= 1/\sqrt{n_k}$, normalized to sum to K. Compresses weight range while preserving relative priorities.

MELD:

- More severe imbalance (up to 37x ratio) required stronger correction — raw inverse frequency with label smoothing epsilon=0.02.

Lesson:

- Combining two imbalance correction mechanisms simultaneously causes compounding over-correction.

Methodology (Models 1-5)

Models 1-2 — Classical text baselines: We followed the standard TF-IDF + classifier design. Key finding: LR (89.7%) outperforms RF (85.6%) -- sparse high-dimensional representations favor linear boundaries.

Model 3 — DistilBERT: We followed the standard DistilBERT fine-tuning design. Key implementation detail: the pretrained CLS token embeddings is reused directly as the text encoder in both multimodal fusion models (models 7-8). AdamW lr=2e-5, 10 epochs, 91.8% accuracy.

Model 4 — CNN from scratch: We followed the standard CNN design of 3 conv blocks, 32->64->128 filters, 48x48 input). Key finding: 55.4% accuracy. Angry class had 0% recall – not an imbalance issue. For example: Ahegao had way fewer samples yet achieves 82% F1. The root cause is feature capacity.

Model 5 — ResNet-18 transfer: We followed the ResNet design from class but implemented it differently: two-phase fine-tuning with differential learning rates.

- Phase 1: freeze backbone, train head only (10 epochs, lr=1e-3)
- Phase 2: unfreeze layer4, differential LRs – backbone lr=1e-4, head lr=1e-3 (20 epochs)
- Result: 79.6% accuracy (+24.2 pts over CNN), Angry: 0% -> 68% F1

Methodology (Models 6-8)

Models 6 — Multimodal Fusion with Unaligned Data: We paired the Twitter text dataset with the OAHEGA images by matching emotion labels. Key finding: served as an important baseline but failed because the text and image did not have the same context. Had an accuracy of 38.5% which was close to random classification.

Model 7 — Multimodal Fusion with CNN Encoder: We switched to the aligned MELD dataset using DistilBERT for text, our CNN from scratch for images, and cross-modal attention for fusion. Key finding: alignment improved performance compared to the unaligned baseline, but the CNN image encoder remained a bottleneck. Achieved 42.2% accuracy, a 3.7% accuracy boost.

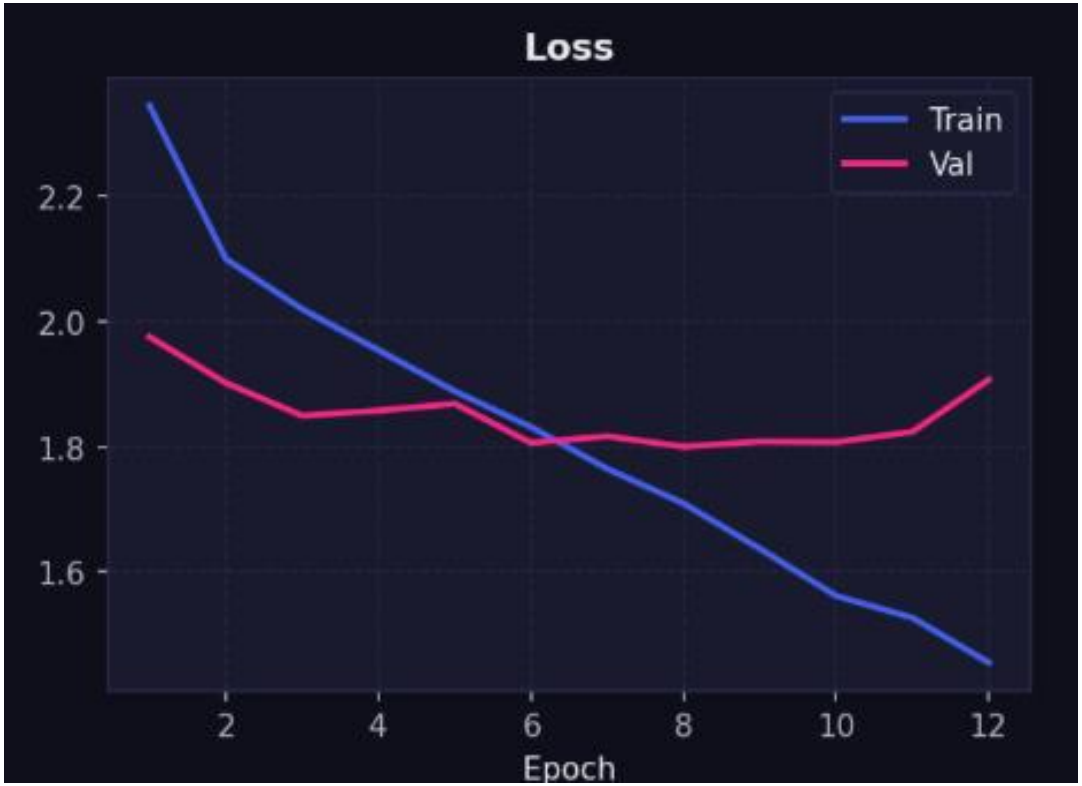
Model 8 — Final System Architecture: We kept the aligned MELD dataset, DistilBERT text encoder, and cross-modal attention fusion, but replaced the CNN with the pretrained ResNet-18 for images. Since the encoders output different vector sizes, a linear layer is applied to produce matching 256-dimensional features. Cross-modal attention lets each modality interact before fusion. The final 512-dimensional vector is then classified into 7 emotion classes. Achieved accuracy of 47.6%, a 5.4 percentage point improvement over model 7.

Results and Findings

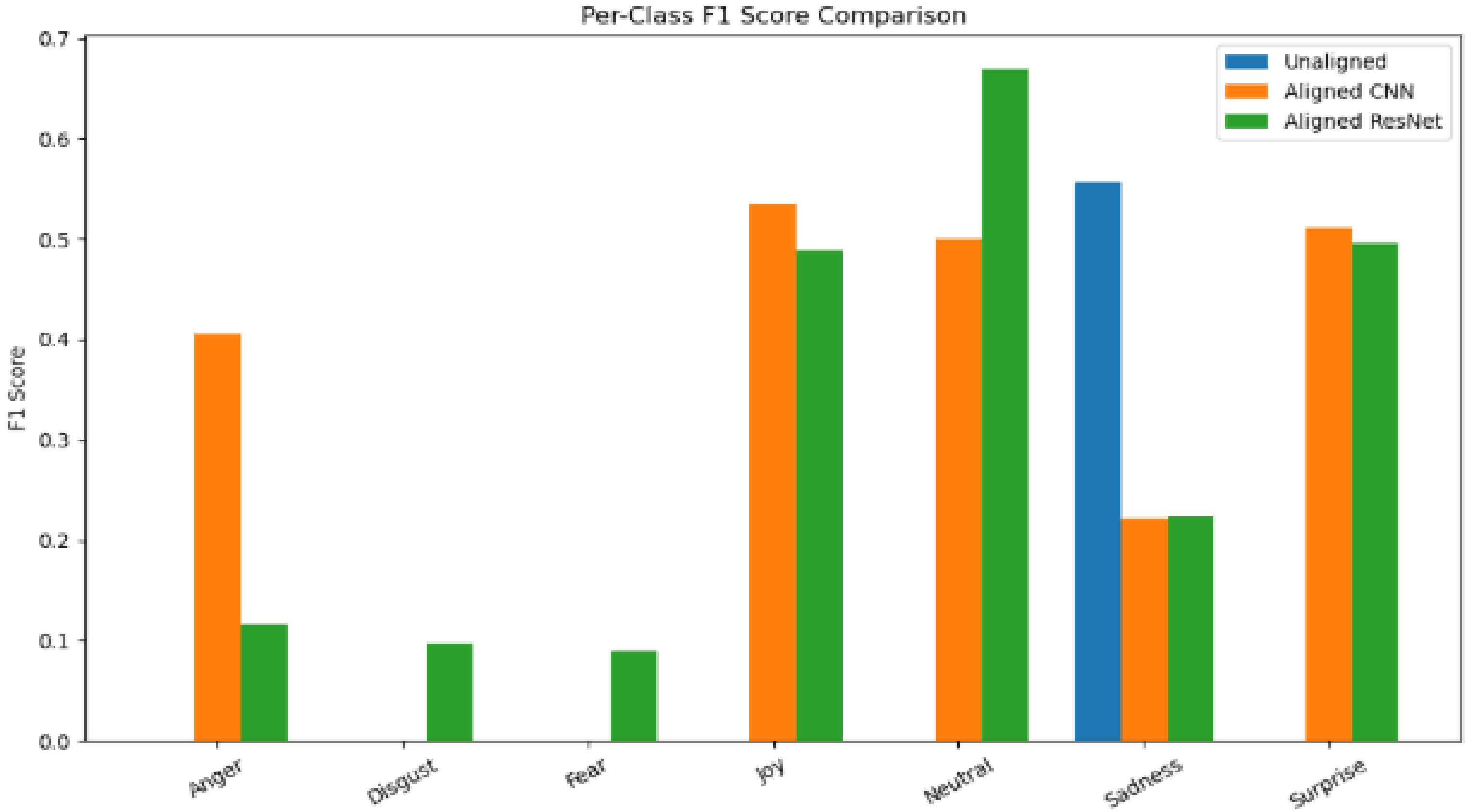
Model	Accuracy	Macro F1	Kappa	ROC-AUC
Random Forest (TF-IDF)	85.6%	—	—	—
Logistic Regression (TF-IDF)	89.7%	—	—	—
DistilBERT (fine-tuned)	91.8%	—	—	—
CNN from scratch	55.4%	0.41	—	—
ResNet-18 (transfer learning)	79.6%	0.80	—	—
Multimodal — unaligned (failed)	38.5%	0.14	0.00	0.50
Multimodal — MELD + CNN	42.2%	0.31	0.29	0.74
Multimodal — MELD + ResNet-18	47.6%	0.31	0.31	0.74

Loss Curve

- Converge through epoch 6
- Early stopping selected epoch 8 checkpoint



Performance of unaligned vs aligned data



Conclusion

- The multimodal model trained on unaligned data failed to learn meaningful emotional expressions.
- Models trained with aligned MELD dataset demonstrated clear improvements in both accuracy and class-level performance.
- However, Multimodal models did not surpass the performance of the best unimodal text model.
- More advanced fusion strategies may be required.

Limitations

- The multimodal models relied on a single frame representation of video data, which may not fully capture dynamics of facial expressions.
- lass imbalance in the datasets posed a significant challenge.
- Despite weighting strategies, minority classes such as disgust and fear remained difficult to classify accurately.

Next steps

- explore more sophisticated multimodal fusion techniques
- Incorporate additional modality – audio

Thank You!

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